

MATH 4360 HOMEWORK ASSIGNMENT 2

DUE ON THURSDAY 7 SEPTEMBER 2023

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Let F be a field and $f(x)$ and $g(x)$ be polynomials in $F[x]$. Set

$$S = \{p(x)f(x) + q(x)g(x) \mid p(x), q(x) \in F[x]\}.$$

- (a) Show that S is an ideal in $F[x]$.
(b) Find the generator of S .

- (2) Let F be a field and $f(x)$ and $g(x)$ be polynomials in $F[x]$. Show that if $f(a) = g(a)$ holds for infinitely many $a \in F$, then one has $f(x) = g(x)$.

- (3) Find all rational roots of the polynomial

$$x^3 + 2x^2 - 5x + 6.$$

- (4) Decide if the polynomial

$$3x^5 + 6x^4 - 54x^2 + 12x - 18$$

is irreducible in $\mathbb{Q}[x]$.